

Woodward's "Explanation, Truth and Structural Realism" II

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June 10, 2026

4. Is the Eliminativist Schema Too Demanding?

- a. A common objection to Schema S: it is too demanding.
 - Specifically, it is impossible to provide a fully exhaustive list of alternative explanations, or we cannot empirically exclude all of them.
- b. Responses:
 - 1) The idea that undermining alternatives is crucial to inductive support is common to many other accounts of confirmation and makes obvious normative sense.
 - E.g. Bayesianism, a probabilistic version of eliminativist ideal; statistical learning theory: "we start with a class of hypotheses and use empirical data to select one hypothesis from the class".
 - 2) We compare explanations involving claims made by effective theories about dependency relations, rather than ontology, within some circumscribed domain, which greatly restricts the set of alternatives and find evidence that discriminates easier.
 - Effective theories: "being approximation to some general theory which at present may be unknown" (p.1)
 - E.g. Newtonian mechanics and gravitational theory within their domains of application: no alternatives exist that are fundamentally different and inconsistent with Newtonian theory and yet compatible with its evidence in its domain of application.
 - Sometimes possible (and even often) to generate a set of possible alternative explanatory theories and search in a systematic way that design observations and experiments that exclude alternative possibilities. E.g. Post Newtonian Formalism which characterizes alternative gravitational theory to GR; causal learning from experimental and causal modeling: algorithms for learning causal relations proceed by systematically generating all causal models that satisfy certain general constraints and

systematically search through alternatives and eliminate them based on statistical evidence.

- There are powerful arguments based on generally accepted evidence and generic theoretical considerations or background knowledge that alternatives to generally accepted theories do not exist. E.g. combination of non-relativistic quantum mechanics, Poincaré invariance, and cluster decomposition lead to quantum field theory.
- Even if formally arguments like the above do not exist, it seems often true that alternative explanatory theories that are consistent with the available evidence do not exist in other fields like neuroscience and biology. It seems that in these cases we have successfully carried out some version of the eliminative schema S – alternatives that have been implicitly/ explicitly eliminated. And it seems true that researchers usually have some sense if there exist unrecognized alternatives.

3) Suppose there are situations in which we are unable to follow the requirement of the eliminative schema S – that there are alternative explanations we have not thought of or unable to rule out on evidential grounds.

- It'd be nice to establish that despite this our explanation is still the correct one!
- But how might this work? The only remaining possibility is through non-empirical signs of truth. But this won't help with the problem of unconceived alternatives, and lack reason to think why non-empirical signs should be indicative of truth.
- So if we are unable to follow the schema, we are not entitled to believe in the truth/truthlikeness of our current explanations.

5. Structure vs Ontology in Explanation

- a. Ian Hacking's "If you can spray them they are real" is more complex than what this slogan suggests.
 - Hacking also writes: "the experimenter is convinced of the reality of entities some of whose causal properties are sufficiently well understood that they can be used to interfere elsewhere in nature"; "Nobody thinks that electrons 'really' are just little

spinning orbs about which you could, with small enough hand, wrap the fingers and find the direction of the spin along the thumb. There is instead a family of causal properties in terms of which gifted experimenters describe and deploy electrons in order to investigate something else, e.g., weak neutral currents and neutral bosons.”

- Hacking’s emphasis not just on manipulating but manipulating one kind of entity to produce changes in another. Success of manipulating taken to be the reason for regarding the first kind of entity as real.
- b. Can extract another lesson from his remarks: relations we are justified in believing are real are the causal relations which we make use of when we manipulate one thing to reliably affect another.
 - Such interference can be reliable even if in the presence of false or incomplete information about the nature of what we are manipulating. E.g. J.J.Thomson’s pudding model of atoms and subjecting electrons to E&M fields allows him to discover the causal properties such as they are negatively charged.
 - This is true in many other cases: we can have mistaken and even radically false beliefs about the nature of the objects we are manipulating, and yet have correct beliefs about the effects of manipulation or relationship between manipulations and other outcomes.
 - The W-account of explanation follows the structural realist picture that it is possible to get the dependency relationship right within its specific domain of application, but get the underlying ontology seriously wrong. **What explains, according to the W-account, has to be limited to claims about dependency relations.**

6. Two Additional Applications of W-account of Explanation

- a. Equivalent Formulations
 - Sometimes philosophers take Heisenberg and Schrödinger "pictures" in quantum mechanics, or general relativity in its standard formulation and its alternative formulation in which Minkowskian spacetime is assumed as background to be two different theories.

- Scientists don't worry which one is the correct one, but treat them as equivalent: Woodward thinks of this attitude as a structuralist stance: it is the dependency relation that explains.

b. False Modeling Assumptions

- Some strands in philosophy of science claim that many scientific models/theories are full of "falsehood" or "drastic distractions".
- There are cases in which models/theories make ontological claims that are not even approximately correct, but the models nonetheless make correct claims about dependency relations and sometimes useful for explanation for that reason.

7. More on Structural Realism: Epistemic vs Ontic Structural Realism

a. Ontic structural realism

- Sometimes claim that only relations but not relata are real – implying that there is relation without relata.
- Woodward thinks this makes structural realism unnecessarily mysterious.
- First, relata that figure in causal or lawful relationships are virtually always properties or quantities like mass, charge, energy. These are not "things" or "entities" in the way that "electrons" or "ether" are. They do change over time in our theories. By contrast, values of entities or quantities have not changed much over time even if ontology changes.
- Second, one can also think that relationships require some relata but we have incomplete understanding of the relata, insofar as with respect to their "fundamental nature", but we don't care that much about their nature as long as we think of dependency relations as doing the explanatory works.

b. Epistemic structural realism

- Some claim that we can have knowledge only about relations but not about relata.
- This too is unnecessarily too strong: we can have reliable information about the properties such as mass, charge, spin.

8. Entity Realism

- a. Entity realism talks of science as yielding knowledge about the nature of unobservables entities.

- Sometimes expressed in terms claims about successful reference – the Putnam/Boyd line that terms in science typically successfully refers, typically supported by a Kripke/Putnam theory of reference, which allows one to successfully refer even if one has false beliefs about the reference – that is, through appropriate causal connection to the entity.
- b. Arguments for entity realism reinforce the significance of structural realism, according to Woodward.
 - Kripke/Putnam theory of reference supports the observation that one can refer to X, distinguish them from other things, and learn about the dependency relations in which they figure even if we do not know much about the underlying nature or our conception changes drastically over time.
- c. Some, such as Worrall, emphasized continuity at the level of mathematics.
 - Woodward believes this is a mistake: mathematics describe independently existing dependency relations.
 - For there are different ways to mathematically describe the same dependency relations, and also that dependency relations in many parts of science are not formulated mathematically.

9. In This “Instrumentalism”?

- Traditionally instrumentalism is taken to be committed to the following four claims:
 - (i) Theories are tools for calculations or making predictions and do not have truth values, at least insofar as what they claim about unobservables.
 - (ii) Relatedly, one can divide theories into claims about observables and unobservables, and the goal of science is to capture the relations among observables correctly.
 - (iii) Theories only make predictions, they do not explain.
 - (iv) It is mistaken to view theories to have literally true or false modal commitments.
- Woodward denies all four claims.

10. The No Miracles Argument(s)

- a. Woodward distinguishes two forms of the no miracles argument(s):
 - The “retail” argument: attempts to establish the truth/near truth of particular theories – that it would be a miracle our science is so successful if they are not nearly true/true.

- The “wholesale” argument: suppose that T satisfies schema S, but an (instrumentalist) researcher does not claim that T is truthlike or explanatory, for the goal of science is not to explain or T commits to things like unobservables which she thinks is unwarranted to believe in. So she doesn’t really claim the success of T is a miracle, but something in the neighborhood: the predictions of T all obtain but no explanation for its success
- b. Remarks on the Implication of Woodward Account for the No-Miracle Arguments:
 - For the first formulation, it’s no miracle if there is an alternative to T incompatible with T and yet make the same predictions; so to argue the truthlikeness of T such that there is no alternative to something like schema S.
 - For the second formulation, the no miracles argument is just an appeal to premise 1 in schema S – that there must be some explanations for T’s successful predictions.

Discussion Questions:

1. Is it true that what explains in theories is **only** the dependency relations they express?
How would Woodward respond to the case of Perrin’s argument for molecular reality that seems to explain the converging value of the measurement of Avogadro’s number? Would he deny the strength of the argument Perrin relies on for molecular reality as explanation?
2. Woodward denies the instrumentalist’s claim that theory does not have a truth-value. But then what is the truth-value of a theory based on Woodward’s notion of “truthlikeness” of a theory? How do you understand the notion of “truthlikeness”?
3. How would Woodward respond to a Duhemian claim according to which theories only represent, but do not explain? Do you still think it is unsatisfactory, as Woodward claims it is, that the success of a theory in making a prediction does not mean the theory is explanatory if we think the theory correctly represents instead?